Assignment Name - Practice Assignment - GCP

Created by - Saleem Shaikh

Batch - 16A

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GIT Repo - https://github.com/saleeminfodevops/herovired.git

Description

Deploy a MERN application using Kubernetes. Make sure the proper documentation along with screenshots of the steps are mentioned in your document and pushed into a GIT repository.

Link to the application:

https://github.com/UnpredictablePrashant/SampleMERNwithMicroservices

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Submission Instructions:

To submit your assignment, please follow these guidelines:

- Ensure that your assignment is fully completed.
- Push your code to a GitHub repository.
- Share the repository link by including it in a text, Word, or PDF file format.

Submit the file/text containing the repository link via Vlearn.

Step 1: Prerequisites & Clone the Repository

1. Install and initialize the Google Cloud CLI (`gcloud`).

Clone the repository to your local machine: [1, 2, 3, 4]

bash

```
git clone
https://github.com/UnpredictablePrashant/SampleMERNwithMicroservices.git
cd SampleMERNwithMicroservices
```

2. Use code with caution.

Step 2: Authenticate and Configure GCP Project

Log in to your GCP account via terminal:

```
gcloud auth login
```

Set your target project ID and default compute region/zone:

```
gcloud config set project heroviredacademics
gcloud config set compute/zone us-central1-c
```

```
learning@inblrlt-saleem:~$ gcloud auth login
Your browser has been opened to visit:

    https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.
2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https
oogleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.go
UI&access_type=offline&code_challenge=vC0xmYqnfQ8zq9XKxGOoBC7yR4jmHS98QSEFSoYGdAI&code_challenge_method=S256

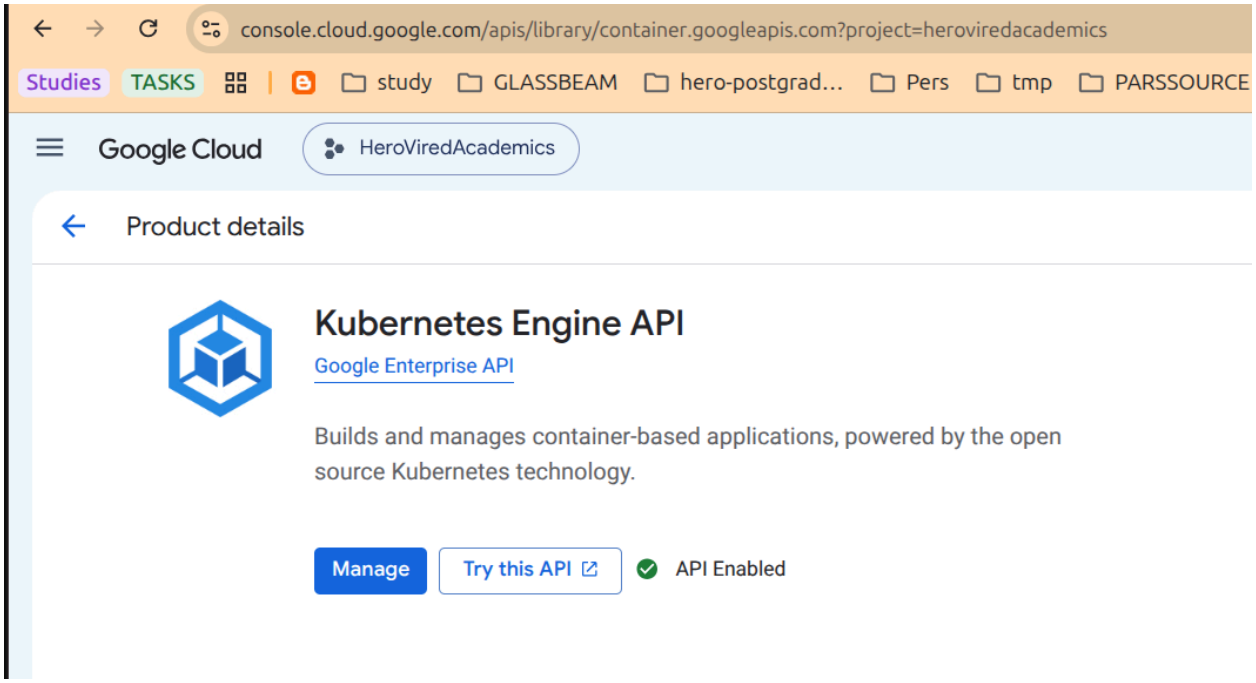

You are now logged in as [saleeminfo@gmail.com].
Your current project is [heroviredacademics].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID


To take a quick anonymous survey, run:
  $ gcloud survey

learning@inblrlt-saleem:~$ gcloud config set compute/zone us-central1-c
Updated property [compute/zone].
```

1. Use code with caution.

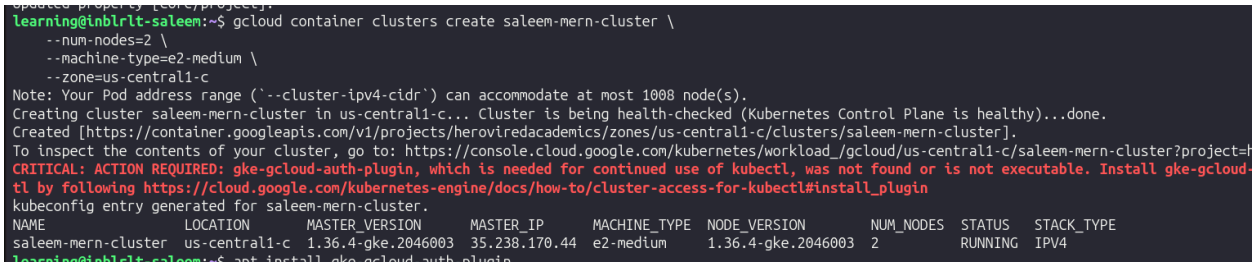
Enable the required Kubernetes Engine API: [1, 2]

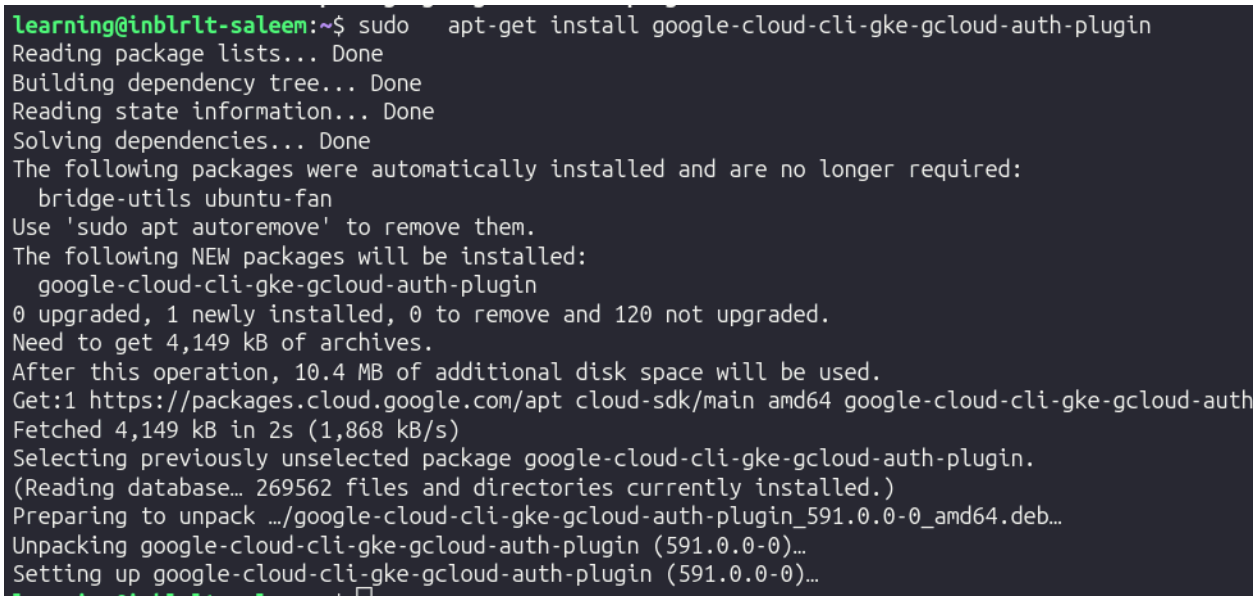


Step 3: Create a GKE Kubernetes Cluster

Create a standard GKE cluster using the `gcloud`

```
gcloud container clusters create saleem-mern-cluster \
    --num-nodes=2 \
    --machine-type=e2-medium \
    --zone=us-central1-c
```





Once the cluster is provisioned, get the authentication credentials so `kubectl` can communicate with your cluster: [1, 2]

```
gcloud container clusters get-credentials mern-cluster
--zone=us-central1-c
```

```
learning@inblrlt-saleem:~$ gcloud container clusters get-credentials saleem-mern-cluster --zone=us-central1-c
Fetching cluster endpoint and auth data.
kubeconfig entry generated for saleem-mern-cluster.
```

Use code with caution.

Verify the node status:

```
kubectl get nodes
```

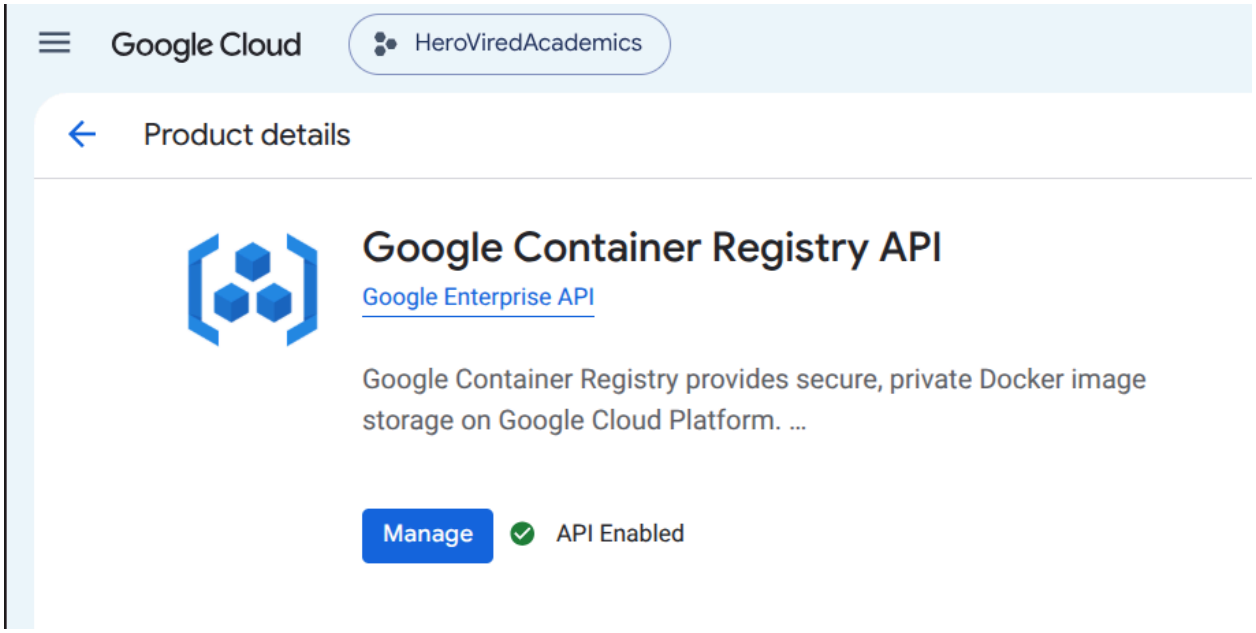
```
learning@inblrlt-saleem:~$ kubectl get nodes
NAME                                                STATUS   ROLES    AGE    VERSION
gke-saleem-mern-cluster-default-pool-9a961138-dbpw   Ready    <none>   4m2s   v1.36.4-gke.2046003
gke-saleem-mern-cluster-default-pool-9a961138-krxr   Ready    <none>   4m3s   v1.36.4-gke.2046003
```

Step 4: Build and Push Docker Images to Artifact Registry / Container Registry

Enable Container Registry in your project:

bash

```
gcloud services enable ://googleapis.com
```

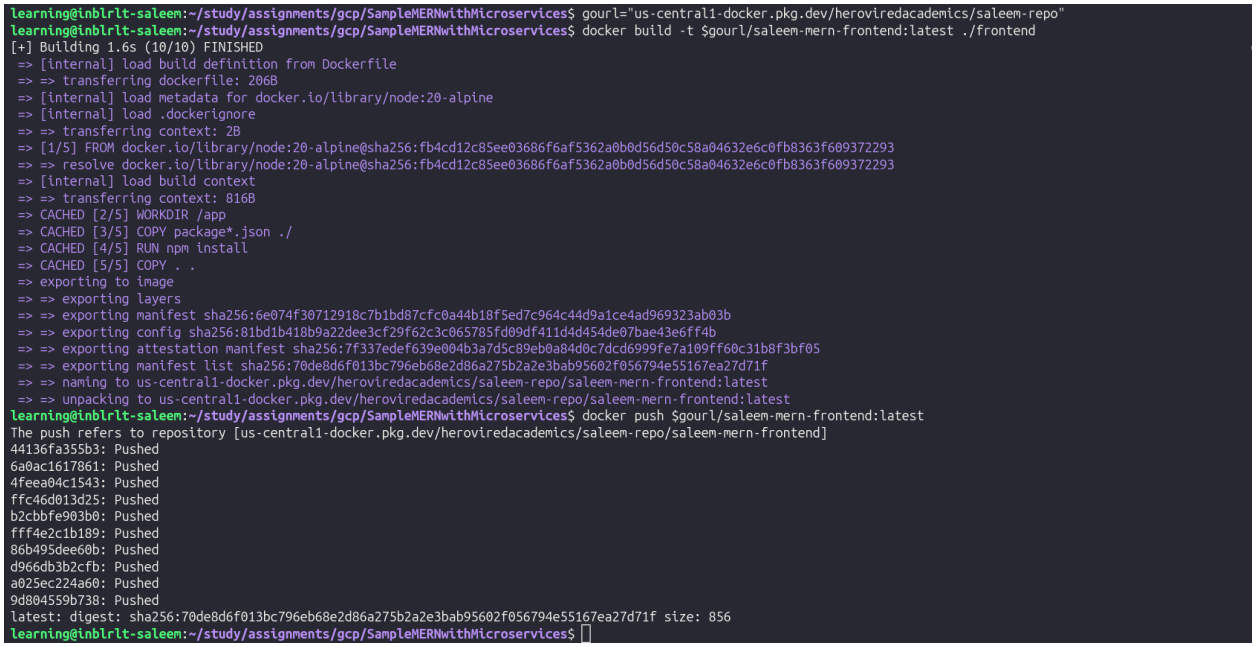


```
gcloud artifacts repositories create saleem-repo
--repository-format=docker --location=us-central1
```

Build and tag your frontend and microservice backend Docker images. For example, configure your Docker to authenticate with GCP:

```
gourl=”us-central1-docker.pkg.dev/heroviredacademics/saleem-repo”

docker build -t $gourl/saleem-mern-frontend:latest ./frontend
docker push $gourl/saleem-mern-frontend:latest
```




```
docker build -t $gourl/saleem-mern-backend-helloservice:latest
./backend/helloService/

docker push $gourl/saleem-mern-backend-helloservice:latest
```

```
Dockerfile  Jenkinsfile  index.js  package-lock.json  package.json
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker build -t $gourl/saleem-mern-backend-helloservice:latest ./backend/helloService/
[+] Building 1.5s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 155B
 => [internal] load metadata for docker.io/library/node:20-alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => [internal] load build context
 => => transferring context: 160B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY package*.json .
 => CACHED [4/5] RUN npm install
 => CACHED [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:a6be2055a89cee0e38d86077e8e9e01f75cb54615d2ecac07f7f412fc4814313
 => => exporting config sha256:5c284c0c8c39f153d356b7206bd5eecf0931169d0fba82e509d20426ebe6e0ee
 => => exporting attestation manifest sha256:798ddfccc6365592fd029c72930e5e2811521603e44fa5f2c50ab71de061d8e7
 => => exporting manifest list sha256:48fd692bda8109466da059622170455a24e517d5d872d9d7e2c142c738c79125
 => => naming to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-helloservice:latest
 => => unpacking to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-helloservice:latest
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker push $gourl/saleem-mern-backend-helloservice:latest
The push refers to repository [us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-helloservice]
4feea04c1543: Layer already exists
44136fa355b3: Already exists
6a0ac1617861: Layer already exists
30a9eb19aaef: Pushed
b2cbbfe903b0: Layer already exists
fff4e2c1b189: Layer already exists
86b495dee60b: Layer already exists
14304f6bd869: Pushed
4950e7e33bc8: Pushed
d55bde13a491: Pushed
latest: digest: sha256:48fd692bda8109466da059622170455a24e517d5d872d9d7e2c142c738c79125 size: 856
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$
```

```
docker build -t $gourl/saleem-mern-backend-profileservice:latest
./backend/profileService/
docker push $gourl/saleem-mern-backend-profileservice:latest
```

```
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker build -t $gourl/saleem-mern-backend-profileservice:latest ./backend/profileService/
[+] Building 2.6s (10/10) FINISHED                                                                                                                  docke
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 284B
 => [internal] load metadata for docker.io/library/node:20-alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
 => [internal] load build context
 => => transferring context: 161B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY package*.json .
 => CACHED [4/5] RUN npm install
 => CACHED [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:0cb3317c695d95ec76e1d8e8f124543f410a3fea95e9e86db74d44ac581fc20c
 => => exporting config sha256:3d22eb8b0a18b5f1ccae1fca7b8afc78b92c96f599e0340948e0d102a9f6c25d
 => => exporting attestation manifest sha256:a56f32826444b9ff8f20d9f833b3d2eb4e14c6977db25f96ca9adb21c78590e3
 => => exporting manifest list sha256:23043893efd938f6678d4418d0f35efddafd1ad2a08abb69ad14f539b33ffb62
 => => naming to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-profileservice:latest
 => => unpacking to us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-profileservice:latest
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$ docker push $gourl/saleem-mern-backend-profileservice:latest
The push refers to repository [us-central1-docker.pkg.dev/heroviredacademics/saleem-repo/saleem-mern-backend-profileservice]
44136fa355b3: Already exists
6a0ac1617861: Layer already exists
70c478d652d0: Pushed
4feea04c1543: Layer already exists
b2cbbfe903b0: Layer already exists
fff4e2c1b189: Layer already exists
86b495dee60b: Layer already exists
29602ebafb10: Pushed
88f9d9b8f348: Pushed
9bca55ab2ac0: Pushed
latest: digest: sha256:23043893efd938f6678d4418d0f35efddafd1ad2a08abb69ad14f539b33ffb62 size: 856
learning@inblrlt-saleem:~/study/assignments/gcp/SampleMERNwithMicroservices$
```

Step 5: Apply Kubernetes Manifests

1. Inspect the Kubernetes deployment and service YAML manifests (`manifest.yml` or files inside the repository). Ensure the image paths inside the YAML point to your pushed GCR/Artifact Registry images. [1]

Deploy the objects to your GKE cluster: [1]

bash

```
kubectl apply -f manifest.yml
```

2. Use code with caution.

Verify that your deployments, pods, and services are running properly: [1]

bash

```
kubectl get deployments
kubectl get pods
kubectl get svc
```

3. Use code with caution.

Step 6: Access Your MERN Application

Check the external IP address or LoadBalancer endpoint assigned to your frontend service: [1, 2]

bash

```
kubectl get svc
```

- Use code with caution.
- Open the external IP or LoadBalancer URL in your browser to interact with the MERN application. [1, 2]

If you'd like, let me know:

- **Which specific database setup** you are using (e.g., MongoDB Atlas vs. in-cluster MongoDB pod)
- **If you run into any build or pod crash errors** (`CrashLoopBackOff`) so we can inspect the logs together.